

An upper bound on the double Roman domination number

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Abstract A *double Roman dominating function* (DRDF) on a graph $G = (V, E)$ is a function $f : V \rightarrow \{0, 1, 2, 3\}$ having the property that if $f(v) = 0$, then vertex v must have at least two neighbors assigned 2 under f or one neighbor w with $f(w) = 3$, and if $f(v) = 1$, then vertex v must have at least one neighbor w with $f(w) \geq 2$. The weight of a DRDF f is the value $f(V) = \sum_{u \in V} f(u)$. The *double Roman domination number* $\gamma_{dR}(G)$ of a graph G is the minimum weight of a DRDF on G . Beeler et al. (Discrete Appl Math 211:23–29, 2016) observed that every connected graph G having minimum degree at least two satisfies the inequality $\gamma_{dR}(G) \leq \frac{6n}{5}$ and posed the question whether this bound can be improved. In this paper, we settle the question and prove that for any connected graph G of order n with minimum degree at least two, $\gamma_{dR}(G) \leq \frac{8n}{7}$.

Keywords Double Roman domination number · Roman domination · Domination

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1 Introduction

Throughout this paper, G is a simple graph with vertex set $V(G)$ and edge set $E(G)$ (briefly V, E). The order $|V|$ of G is denoted by $n = n(G)$. For every vertex $v \in V(G)$, the *open neighborhood* of v is the set $N_G(v) = N(v) = \{u \in V(G) \mid uv \in E(G)\}$ and its *closed neighborhood* is the set $N_G[v] = N[v] = N(v) \cup \{v\}$. The *degree* of a vertex $v \in V$ is $\deg(v) = d(v) = |N(v)|$. The *minimum degree* and the *maximum degree* of a graph G are denoted by $\delta = \delta(G)$ and $\Delta = \Delta(G)$, respectively. A set S is an *independent set* if the induced subgraph $G[S]$ is empty. A path $P = u_0 u_1 \dots u_k$ ($k \geq 1$) in G is called a pendant path if $d_{u_0} \geq 3$, $d_{u_k} = 1$ and the degree of any other vertex of the path is 2.

A set $S \subseteq V$ in a graph G is called a *dominating set* if every vertex of G is either in S or adjacent to a vertex of S . The *domination number* $\gamma(G)$ equals the minimum cardinality of a dominating set in G . The domination and its variations have been attracted considerable attention and surveyed in two books Haynes et al. (1998a, b).

A *double Roman dominating function* (DRDF) on a graph G is a function $f : V \rightarrow \{0, 1, 2, 3\}$ having the property that if $f(v) = 0$, then vertex v must have at least two neighbors assigned 2 under f or one neighbor w with $f(w) = 3$, and if $f(v) = 1$, then vertex v must have at least one neighbor w with $f(w) \geq 2$. The weight of a DRDF is the value $f(V) = \sum_{u \in V} f(u)$. The *double Roman domination number* $\gamma_{dR}(G)$ of a graph G is the minimum weight of a DRDF on G . For a DRDF f , let $V_i = \{v \in V \mid f(v) = i\}$ for $i = 0, 1, 2, 3$. Since these four sets determine f , we can equivalently write $f = (V_0, V_1, V_2, V_3)$ (or $f = (V_0^f, V_1^f, V_2^f, V_3^f)$ to refer f). We note that $\omega(f) = |V_1| + 2|V_2| + 3|V_3|$.

The concept of double Roman domination in graphs was introduced by Beeler, Haynes and Hedetniemi in Beeler et al. (2016) and has been studied in Abdollahzadeh Ahangar et al. (2017a, b; 2018), Volkmann (2018a, b) and Zhang et al. (2018). In Beeler et al. (2016), it was proved that for any connected graph G of order $n \geq 3$, $\gamma_{dR}(G) \leq \frac{5n}{4}$ and all extremal graphs are characterized. Abdollahzadeh Ahangar et al. (2017b) proved that every connected graph G having minimum degree at least 3, satisfies the inequality $\gamma_{dR}(G) \leq n(G)$.

Here we consider connected graphs with minimum degree at least two. A theorem of McCuaig and Shepherd (1989) says that with the exception of seven graphs illustrated in Fig. 1, every connected graph G having minimum degree at least two satisfies the inequality, $\gamma(G) \leq \frac{2n}{5}$. Using this result, Beeler et al. (2016) observed that every connected graph having minimum degree 2, satisfies the inequality $\gamma_{dR}(G) \leq \frac{6n}{5}$ and posed the question whether this bound can be improved.

In this paper, we settle the question and prove that for any connected graph G of order n with minimum degree at least two $\gamma_{dR}(G) \leq \frac{8n}{7}$.

We make use of the following results in this paper.

Proposition A (Beeler et al. 2016) *In a double Roman dominating function of weight $\gamma_{dR}(G)$, no vertex needs to be assigned the value 1.*

By Proposition A, in what follows, when we consider a γ_{dR} -function of a graph G we assume no vertex assigned the value 1.

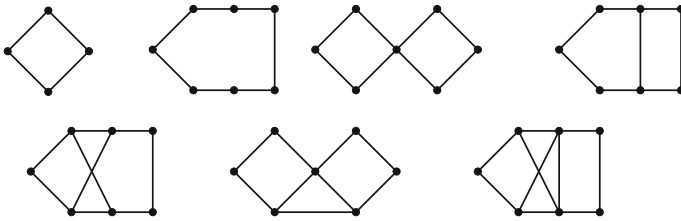


Fig. 1 Graphs with domination number greater than $\frac{2|V(G)|}{5}$

Proposition B (Jafari Rad and Rahbani 2018) *For any connected graph G of order n with maximum degree Δ , $\gamma_{dR}(G) \leq 2n - 2\Delta + 1$.*

Proposition C (Abdollahzadeh Ahangar et al. 2017b) *For $n \geq 1$, $\gamma_{dR}(P_n) = \begin{cases} n & \text{if } n \equiv 0 \pmod{3}, \\ n + 1 & \text{if } n \equiv 1, 2 \pmod{3}. \end{cases}$*

Proposition D (Abdollahzadeh Ahangar et al. 2017b) *For $n \geq 3$, $\gamma_{dR}(C_n) = \begin{cases} n & \text{if } n \equiv 0, 2, 3, 4 \pmod{6} \\ n + 1 & \text{if } n \equiv 1, 5 \pmod{6}. \end{cases}$*

2 Main results

In this section we establish an upper bound on the double Roman domination number of graphs with minimum degree 2. First we introduce two families of graphs.

For integers $m \geq 3$ and $k \geq 1$, let $C_{m,k}$ be the graph obtained from a cycle $C_m = (x_1x_2 \dots x_m)$ by adding a pendant path $x_1y_1y_2 \dots y_k$.

Lemma 1 *For integers $m \geq 3$ and $k \geq 1$ with $m + k \geq 4$, $\gamma_{dR}(C_{m,k}) \leq \frac{8(m+k)}{7}$.*

Proof If $m + k \leq 6$, then clearly $\gamma_{dR}(C_{m,k}) \leq m + k < \frac{8(m+k)}{7}$. Let $m + k \geq 7$. Since $C_{m,k}$ has a Hamiltonian path and since adding an edge cannot increase the double Roman domination number, we deduce from Proposition C that $\gamma_{dR}(C_{m,k}) \leq m + k + 1 \leq \frac{8(m+k)}{7}$. $\square$

Let $\mathcal{F}_1$ be the family of all connected multigraphs without loops and with minimum degree at least 3. Assume that $\mathcal{F}$ is the family of all graphs obtained from some graph in $\mathcal{F}_1$ by subdividing any edge at least once and at most five times. Note that any graph in $\mathcal{F}$ has order at least 5.

Lemma 2 *If $G \in \mathcal{F}$, then there exists a double Roman dominating function $f = (V_0, \emptyset, V_2, V_3)$ of G such that $\omega(f) \leq \frac{8n(G)}{7}$ and f assigns a positive value to every vertex of degree at least 3.*

Proof Let $G \in \mathcal{F}$ be a graph of order n . The proof is by induction on n . The result is immediate for $n \leq 6$. Suppose $n \geq 7$ and let the result hold for all graphs in $\mathcal{F}$ of order less than n . Let $G \in \mathcal{F}$ be a graph of order $n \geq 7$. Suppose that A denotes the set of vertices of degree at least 3 in G , and let $B = V(G) - A$. A path P of G is called maximal if $V(P) \subseteq B$ and each end-vertex of P is adjacent to a vertex

of A . For each $i \geq 1$, let $\mathcal{P}_i = \{P \mid P \text{ is a maximal path with } |V(P)| = i\}$. Let $\mathcal{P} = \bigcup_{i \geq 1} \mathcal{P}_i$. Note that $A \cup \bigcup_{P \in \mathcal{P}} V(P)$ is a partition of $V(G)$. For $P \in \mathcal{P}$, let $X_P = \{u \in A \mid u \text{ is adjacent to an end-vertex of } P\}$. Then $A = \bigcup_{P \in \mathcal{P}} X_P$ and since G is obtained from some multigraph without loops in $\mathcal{F}_1$ by subdividing all of its edges at least once, we have $|X_P| = 2$ for each $P \in \mathcal{P}$. Hence $|A| \geq 2$. First let $\mathcal{P}_3 \cup \mathcal{P}_5 \neq \emptyset$. Suppose $P = x_1 \dots x_{2k+1} \in \mathcal{P}_{2k+1}$ where $k \in \{1, 2\}$ and let $X_P = \{a_1, a_2\}$ where $a_1 x_1, a_2 x_{2k+1} \in E(G)$. Assume that $G' = (G - (V(P) - \{x_{k+1}\})) + \{a_1 x_{k+1}, a_2 x_{k+1}\}$. Clearly, $G' \in \mathcal{F}$. By the induction hypothesis, there exists a double Roman dominating function $f = (V_0, \emptyset, V_2, V_3)$ of G' such that $a_1, a_2 \in V_2 \cup V_3$, and $\omega(f) \leq \frac{8(n-2k)}{7}$. It follows that $f(x_{k+1}) = 0$. If $k = 1$, then the function $g : V(G) \rightarrow \{0, 1, 2, 3\}$ defined by $g(x_2) = 2, g(x_1) = g(x_3) = 0$ and $g(x) = f(x)$ otherwise, is a DRDF of G such that g assigns a positive value to every vertex of degree at least 3, and

$$\omega(g) = \omega(f) + 2 \leq \frac{8(n-2)}{7} + 2 < \frac{8n}{7}.$$

If $k = 2$, then the function $g : V(G) \rightarrow \{0, 1, 2, 3\}$ defined by $g(x_2) = g(x_4) = 2, g(x_1) = g(x_3) = g(x_5) = 0$ and $g(x) = f(x)$ otherwise, is a DRDF of G such that g assigns a positive value to every vertex of degree at least 3, and

$$\omega(g) = \omega(f) + 4 \leq \frac{8(n-4)}{7} + 4 < \frac{8n}{7}.$$

Assume now that $\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2 \cup \mathcal{P}_4$. Note that $n = |A| + m_1 + 2m_2 + 4m_4$ and $m_1 + m_2 + m_4 \geq 3$, where $m_i = |\mathcal{P}_i|$ for $i \in \{1, 2, 4\}$. If $|A| = 2$, then let $\mathcal{P}_4 = \{v_1^i v_2^i v_3^i v_4^i \mid 1 \leq i \leq m_4\}$ and define the function $g : V(G) \rightarrow \{0, 1, 2, 3\}$ by $g(x) = g(v_2^i) = 3$ for each $x \in A$ and each $1 \leq i \leq m_4$, and $g(x) = 0$ otherwise. Clearly, g is a DRDF of G such that g assigns a positive value to every vertex of degree at least 3, and

$$\omega(g) \leq 3|A| + 3m_4 \leq \frac{8n}{7}.$$

Henceforth, we assume $|A| \geq 3$. Let u be a vertex in A such that $\deg(u) = \max\{\deg(x) \mid x \in A\}$. We consider the following cases.

Case 1 u is adjacent to two maximal paths $P_1 \in \mathcal{P}_2$ and $P_2 \in \mathcal{P}_4$.

Let $P_1 = x_1 x_2$ and $P_2 = y_1 y_2 y_3 y_4$ and let $\{ux_1, uy_1, a_1 x_2, a_2 y_4\} \subseteq E(G)$ where $a_1, a_2 \in A$. Assume that G' is the graph obtained from G by removing the vertices u, y_1, y_2 and joining y_3 to each vertex $z \in N_G(u) - \{y_1\}$. Clearly, $G' \in \mathcal{F}$. By the induction hypothesis, there exists a double Roman dominating function $f = (V_0, \emptyset, V_2, V_3)$ of G' such that f assigns a positive value to every vertex of degree at least 3, and $\omega(f) \leq \frac{8(n-3)}{7}$. In particular, $f(y_3) \geq 2$ and $f(a_1) \geq 2$. To double Roman dominate the vertices x_1, x_2 , we must have $f(y_3) + f(a_1) + f(x_1) + f(x_2) \geq 6$. Without loss of generality, we may assume that $f(y_3) = f(a_1) = 3$. Define the function $g : V(G) \rightarrow \{0, 1, 2, 3\}$ by $g(u) = 3, g(y_1) = g(y_2) = 0$ and $g(x) = f(x)$ otherwise. Clearly, g is a DRDF of G such that g assigns a positive value to every vertex of degree at least 3, and

$$\omega(g) = \omega(f) + 3 \leq \frac{8(n-3)}{7} + 3 < \frac{8n}{7}.$$

Case 2 u is adjacent to two paths $P_1, P_2 \in \mathcal{P}_2$.

Let $P_1 = x_1x_2$ and $P_2 = y_1y_2$ be two maximal paths in $\mathcal{P}_2$ and let $\{ux_1, uy_1, ax_2, by_2\} \subseteq E(G)$ where $a, b \in A$. First let $a \neq b$. Assume that G' is the graph obtained from G by removing the vertices x_1, u, y_1 and joining x_2 to y_2 and joining every vertex x in $N(u) - \{x_1, y_1\}$ to either a or b provided a or b is not adjacent to the end-vertex of the maximal path containing x . Clearly, $G' \in \mathcal{F}$. By the induction hypothesis, there exists a double Roman dominating function $f = (V_0, \emptyset, V_2, V_3)$ of G' such that f assigns a positive value to every vertex of degree at least 3, and $\omega(f) \leq \frac{8(n-3)}{7}$. As in Case 1, we may assume that $f(a) = f(b) = 3$. Define the function $g : V(G) \rightarrow \{0, 1, 2, 3\}$ by $g(u) = 3, g(x_1) = g(y_1) = 0$ and $g(x) = f(x)$ otherwise. Clearly, g is a DRDF of G such that g assigns a positive value to every vertex of degree at least 3, and

$$\omega(g) = \omega(f) + 3 \leq \frac{8(n-3)}{7} + 3 < \frac{8n}{7}.$$

Now let $a = b$. Suppose G' is the graph obtained from $G - x_2$ by adding the edge x_1a . Clearly, $G' \in \mathcal{F}$ and by the induction hypothesis, there exists a double Roman dominating function $f = (V_0, \emptyset, V_2, V_3)$ of G' such that f assigns a positive value to every vertex of degree at least 3, and $\omega(f) \leq \frac{8(n-1)}{7}$. As in Case 1, we may assume that $f(a) = f(b) = 3$. Then the function $g : V(G) \rightarrow \{0, 1, 2, 3\}$ defined by $g(x_2) = 0$ and $g(x) = f(x)$ otherwise, is a DRDF of G such that g assigns a positive value to every vertex of degree at least 3, and

$$\omega(g) = \omega(f) \leq \frac{8(n-1)}{7} < \frac{8n}{7}.$$

Case 3 u is adjacent to two paths $P_1, P_2 \in \mathcal{P}_4$.

Considering Case 1, we may assume that u is not adjacent to any maximal path in $\mathcal{P}_2$. Let $P_1 = x_1x_2x_3x_4$ and $P_2 = y_1y_2y_3y_4$ and let $\{ux_1, uy_1, ax_4, by_4\} \subseteq E(G)$ where $a, b \in A$. Consider the following subcases.

Subcase 3.1 $a \neq b$.

Assume that G' is the graph obtained by removing the vertices $x_3, x_2, x_1, u, y_1, y_2, y_3, y_4$ and joining b to x_4 and joining any vertex in $x \in N_G(u) - \{x_1, y_1\}$ to either a or b provided a or b is not adjacent to the end-vertex of the maximal path containing x . Clearly, $G' \in \mathcal{F}$. By the induction hypothesis, there exists a double Roman dominating function $f = (V_0, \emptyset, V_2, V_3)$ of G' such that f assigns a positive value to every vertex of degree at least 3, and $\omega(f) \leq \frac{8(n-8)}{7}$. Define the function $g : V(G) \rightarrow \{0, 1, 2, 3\}$ by $g(u) = g(y_3) = g(x_3) = 3, g(y_1) = g(y_2) = g(y_4) = g(x_1) = g(x_2) = 0$ and $g(x) = f(x)$ otherwise. Clearly, g is a DRDF of G such that g assigns a positive value to every vertex of degree at least 3, and

$$\omega(g) = \omega(f) + 9 \leq \frac{8(n-8)}{7} + 9 < \frac{8n}{7}.$$

Subcase 3.2 $a = b$.

Suppose that u is adjacent to a path $P_3 = z_1 z_2 z_3 z_4 \in \mathcal{P}_4 - \{P_1, P_2\}$ such that $\{u z_1, a z_4\} \subseteq E(G)$. Assume that G' is the graph obtained by removing the vertices $x_3, x_2, x_1, u, y_1, y_2, z_1, z_2$ and joining z_3 to y_3, x_4 and any vertex in $N_G(u) - \{x_1, y_1, z_1\}$. It is easy to see that $G' \in \mathcal{F}$. By the induction hypothesis, there exists a double Roman dominating function $f = (V_0, \emptyset, V_2, V_3)$ of G' such that f assigns a positive value to every vertex of degree at least 3, and $\omega(f) \leq \frac{8(n-8)}{7}$. As Case 1, we may assume that $f(a) = f(z_3) = 3$. Define the function $g : V(G) \rightarrow \{0, 1, 2, 3\}$ by $g(u) = g(x_2) = g(y_2) = 3, g(x_1) = g(y_1) = g(x_3) = g(z_1) = g(z_2) = 0$ and $g(x) = f(x)$ otherwise. Clearly, g is a DRDF of G such that g assigns a positive value to every vertex of degree at least 3, and

$$\omega(g) = \omega(f) + 9 \leq \frac{8(n-8)}{7} + 9 \leq \frac{8n}{7}.$$

Henceforth, we may assume that each vertex in $N(u) - \{x_1, y_1\}$ belongs to a maximal path in $\mathcal{P}_1$. First, let there exist a path $P_3 = z \in \mathcal{P}_1$ such that $zb \in E(G)$ where $b \in A - \{u, a\}$. Assume that G' is the graph obtained by removing the vertices $x_3, x_2, x_1, u, y_1, y_2, y_3$ and joining b to y_4 , joining z to x_4 and joining any vertex in $x \in N_G(u) - \{x_1, y_1, z\}$ to either a or b provided a or b is not adjacent to an end-vertex of the maximal path containing x . Clearly, $G' \in \mathcal{F}$. By the induction hypothesis, there exists a double Roman dominating function $f = (V_0, \emptyset, V_2, V_3)$ of G' such that f assigns a positive value to every vertex of degree at least 3, and $\omega(f) \leq \frac{8(n-7)}{7}$. As in Case 1, we may assume that $f(a) = f(b) = 3$. Define the function $g : V(G) \rightarrow \{0, 1, 2, 3\}$ by $g(y_2) = g(x_2) = 3, g(u) = 2, g(y_1) = g(y_3) = g(x_1) = g(x_3) = 0$ and $g(x) = f(x)$ otherwise. Clearly, g is a DRDF of G such that g assigns a positive value to every vertex of degree at least 3, and

$$\omega(g) = \omega(f) + 8 \leq \frac{8(n-7)}{7} + 8 \leq \frac{8n}{7}.$$

Therefore, we assume that each maximal path with an end-vertex adjacent to u is between u and a . We conclude from the choice of u and the connectedness of G that $A = \{u, a\}$ which is a contradiction.

Case 4 u is adjacent to a path $P_1 \in \mathcal{P}_4$ and to two paths $P_2, P_3 \in \mathcal{P}_1$.

Let $P_1 = x_1 x_2 x_3 x_4$ such that $\{u x_1, a x_4\} \subseteq E(G)$ where $a \in A$. By Cases 1, 2 and 3, we may assume that the other neighbors of u belong to maximal paths in $\mathcal{P}_1$. Assume that G' is the graph obtained from $G - \{u, x_1\}$ by joining x_2 to every vertex in $N(u) - \{x_1\}$. Obviously $G' \in \mathcal{F}$. By the induction hypothesis, there exists a double Roman dominating function $f = (V_0, \emptyset, V_2, V_3)$ of G' such that f assigns a positive value to every vertex of degree at least 3, and $\omega(f) \leq \frac{8(n-2)}{7}$. The function $g : V(G) \rightarrow \{0, 1, 2, 3\}$ defined by $g(u) = 2, g(x_1) = 0$ and $g(x) = f(x)$ otherwise, is a DRDF of G such that g assigns a positive value to every vertex of degree at least 3, and

$$\omega(g) = \omega(f) + 2 \leq \frac{8(n-2)}{7} + 2 < \frac{8n}{7}.$$

Considering the cases above, we may assume that $\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2$ and that each vertex in A is adjacent to at most one maximal path in $\mathcal{P}_2$. Since $\deg(a) \geq 3$ for each $a \in A$, we deduce that each vertex in A is adjacent to at least two maximal paths in $\mathcal{P}_1$. Counting the edges between A and $\cup_{P \in \mathcal{P}_1} V(P)$ implies that $|A| \leq m_1$. Let $A' = \{u \in A \mid u \text{ is adjacent to an end-vertex of a maximal path in } \mathcal{P}_2\}$ and $A'' = A - A'$. Counting the edges between A' and $\cup_{P \in \mathcal{P}_2} V(P)$ yields $|A'| \leq 2m_2$. Define the function $g : V(G) \rightarrow \{0, 1, 2, 3\}$ by $g(x) = 3$ for $x \in A'$, $g(x) = 2$ for $x \in A''$ and $g(x) = 0$ otherwise. It is easy to see that g is a DRDF of G that assigns a positive value to every vertex of degree at least 3 and

$$\omega(g) \leq 3|A'| + 2|A''| = 2|A| + |A'| \leq |A| + m_1 + 2m_2 = n < \frac{8n}{7}.$$

This completes the proof. $\square$

Theorem 3 *Let $G \neq C_5$ be a simple graph of order n with $\delta(G) \geq 2$. Then*

$$\gamma_{dR}(G) \leq \frac{8n}{7}.$$

This bound is sharp for C_7 .

Proof The proof is by induction on n . If $n = 5$ or $n = 6$ and $\Delta(G) \neq 3$, then the result follows from Propositions B or D. If $n = 6$ and $\Delta = 3$, then it is easy to check that $\gamma_{dR}(G) \leq n < \frac{8n}{7}$. Suppose $n \geq 7$ and the result holds for all graph G of order less than n with $\delta(G) \geq 2$. Let G be a graph of order $n \geq 7$ and $\delta(G) \geq 2$. Since $\gamma_{dR}(G) \leq \gamma_{dR}(G - e)$ for every edge $e \in E(G)$, we may assume that $|E(G)|$ is as small as possible. If G is disconnected and $G_1, \dots, G_t$ are the components of G , then it follows from the induction hypothesis that $\gamma_{dR}(G_i) \leq \frac{8|V(G_i)|}{7}$ for each i and so

$$\gamma_{dR}(G) = \sum_{i=1}^t \gamma_{dR}(G_i) \leq \sum_{i=1}^t \frac{8|V(G_i)|}{7} = \frac{8n}{7}.$$

Let G be connected. If $\Delta(G) = 2$, then the result follows by Proposition D. Assume that $\Delta(G) \geq 3$. Let $A = \{v \in V(G) \mid \deg(v) \geq 3\}$ and $B = V(G) - A$. Since $|E(G)|$ is as small as possible, A is independent. Consider the maximal paths and associated notations defined in the proof of Lemma 2. Note that $A = \bigcup_{P \in \mathcal{P}} X_P$, $A \cup \bigcup_{P \in \mathcal{P}} V(P)$ is a partition of $V(G)$ and $1 \leq |X_P| \leq 2$ for each $P \in \mathcal{P}$. Assume first that there exists a maximal path P such that $\delta(G - V(P)) \leq 1$. This implies that $|X_P| = 1$ and since G is simple we have $|V(P)| \geq 2$. Suppose that $X_P = \{a\}$ and let $N_G(a) - V(P) = \{b\}$. Then there exists the unique maximal path P' such that b is an end-vertex of P' . Let $G' = G - (V(P) \cup V(P') \cup \{a\})$. Then $\delta(G') \geq 2$ and by the induction hypothesis we have

$$\gamma_{dR}(G') \leq \frac{8|V(G')|}{7}.$$

On the other hand, since $G'' = G[V(P) \cup V(P') \cup \{a\}] \cong C_{|V(P)|+1, |V(P')|}$, we have

$$\gamma_{dR}(G[V(P) \cup V(P') \cup \{a\}]) \leq \frac{8|V(P) \cup V(P') \cup \{a\}|}{7},$$

by Lemma 1. If f is a $\gamma_{dR}(G')$ -function and g is a $\gamma_{dR}(G'')$ -function, then the function h defined on $V(G)$ by $h(x) = f(x)$ for $x \in V(G')$ and $h(x) = g(x)$ for $x \in V(G'')$, is a DRDF of G and hence

$$\begin{aligned} \gamma_{dR}(G) &\leq \gamma_{dR}(G') + \gamma_{dR}(G[V(P) \cup V(P') \cup \{a\}]) \\ &\leq \frac{8|V(G')|}{7} + \frac{8|V(P) \cup V(P') \cup \{a\}|}{7} \\ &= \frac{8n}{7}. \end{aligned}$$

Assume that $\delta(G - V(P)) \geq 2$ for each maximal path $P \in \mathcal{P}$. It follows that $|X_P| = 2$ for each maximal path $P \in \mathcal{P}$. First let $\mathcal{P} - (\mathcal{P}_1 \cup \mathcal{P}_2 \cup \mathcal{P}_3 \cup \mathcal{P}_4 \cup \mathcal{P}_5) \neq \emptyset$. Suppose $Q \in \mathcal{P} - (\mathcal{P}_1 \cup \mathcal{P}_2 \cup \mathcal{P}_3 \cup \mathcal{P}_4 \cup \mathcal{P}_5)$ and $G' = G - V(Q)$. By Proposition C and the induction hypothesis we have $\gamma_{dR}(Q) \leq \frac{8|V(Q)|}{7}$ and $\gamma_{dR}(G') \leq \frac{8|V(G')|}{7}$. If f is a $\gamma_{dR}(G')$ -function and g is a $\gamma_{dR}(Q)$ -function, then the function h defined on $V(G)$ by $h(x) = f(x)$ for $x \in V(G')$ and $h(x) = g(x)$ for $x \in V(Q)$, is a DRDF of G and hence $\gamma_{dR}(G) \leq \gamma_{dR}(G') + \gamma_{dR}(Q) \leq \frac{8n}{7}$. Now let $\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2 \cup \mathcal{P}_3 \cup \mathcal{P}_4 \cup \mathcal{P}_5$. Then $G \in \mathcal{F}$ and the result follows from Lemma 2. This completes the proof. $\square$

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